Day 4 Calculating SVD by hand

Mathmatical Assignment:

Problem 1: Finding the singular values (Σ)

given 3X2 matrix

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix}$$

TaskA: Calci matrix product $A^T.A$. What are the dimension of new matrix? Notice that is perfectly symmetric.

TaskB: Find the characteristics of polynomial of $A^T.A$ and solve for its eigen values λ_1 and λ_2 (assume $\lambda_1 > \lambda_2$).

TaskC: The singular values of A are the square roots of the non zero eigenvalues of $A^T.A$. Calci the singular value σ_1, σ_2.

Task D: Construct the 3X2 singular values Σ.

Solution:

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix} \qquad A^T = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix}$$

CS CamScanner

$A.A^T = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$

The dimension : 2X2

<u>TaskB:</u> Since $A.A^T$ is a diagonal matrix, its eigenvalues are simply entire on the main diagonal.

$\lambda_1 = 3, \quad \lambda_2 = 2. \quad \lambda_1 > \lambda_2$

<u>TaskC:</u>

$\sigma_1 = \sqrt{\lambda_1} = \sqrt{3} \qquad \sigma_2 = \sqrt{\lambda_2} = \sqrt{2}$

<u>TaskD:</u> 3X2 singular value $\Sigma = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{3} \\ 0 & 0 \end{bmatrix}$

<u>problem 2:</u> Finding the right singular vector (V)

<u>TaskA:</u> For the eigenvalue λ_1, find the corresponding eigenvector v_1. Normalize it so its length 1. ($\|v\| = 1$)

<u>TaskB:</u> . . . λ_2, ... v_2.

<u>TaskC:</u> Construct 2X2 orthogonal matrix V using v_1 and v_2 as columns. Then write out V^T.

CamScanner

Solution:

TASK A: $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$, $AA^T = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$

v_1 for $\lambda_1 = 3$:

$$(A.A^T - \lambda_1 I)v = 0$$

$$\Rightarrow \left(\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} - 3\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) v = 0$$

$$\Rightarrow \left(\begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} \right) v = 0$$

$$\Rightarrow \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This means $-x = 0$, $x = 0$. The value of y can be anything, so we chose $y = 1$ for a unit vector

$$v_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

TASK B: v_2 for $\lambda_2 = 2$:

$$(A.A^T - 2I) = 0$$

$\Rightarrow \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ This means $y=0$. The value of x can be anything. So we chose $x=1$ for unit vector

$v_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Task C:

$$V = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \longrightarrow V^T = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

problem 3: Finding the left singular vectors (U).

$$Av_i = \sigma_i u_i \Rightarrow u_i = \frac{1}{\sigma_i} Av_i$$

Task A: Use the formula above to calculate u_1 using your previously found σ_1 and v_1

Task B: Calc u_2 using σ_2 and v_2.

Solution:

$$u_1 = \frac{1}{\sigma_1} A v_1$$

Task A:

$$= \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

Task B

$$u_2 = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{bmatrix}.$$

CamScanner